

ASSIGNMENT ON SPECIAL OPERATORS

MySQL Query Checking — Based on the queries and outputs provided

Important: Queries are preserved as written. Incorrect queries are not corrected; the issue is only explained in the comment below them.

1. LIST ALL THE EMPLOYEES WHOSE COMMISSION IS NULL.

Query:

```
Select * from emp where comm IS null;
```

✓ CORRECT

-- COMMENT: Your query correctly checks employees whose COMM value is NULL.

2. LIST ALL THE EMPLOYEES WHO DON'T HAVE A REPORTING MANAGER.

Query:

```
Select * from emp where mgr is null;
```

✓ CORRECT

-- COMMENT: This is the query you executed in the MySQL log. It correctly checks for a NULL reporting manager.

3. LIST ALL THE SALESMAN IN DEPT 30.

Query:

```
select * from emp where job='SALESMAN' and deptno=30;
```

✓ CORRECT

-- COMMENT: The query checks both SALESMAN and department 30.

4. LIST ALL THE SALESMAN IN DEPT NUMBER 30 AND HAVING SALARY GREATER THAN 1500.

Query:

```
select * from emp where deptno=30 and sal>1500;
```

✗ WRONG / ISSUE

-- COMMENT: COMMENT: The query checks department 30 and salary > 1500, but it does not check that JOB is SALESMAN. Therefore BLAKE (MANAGER) is also returned.

5. LIST ALL THE EMPLOYEES WHOSE NAME STARTS WITH 'S' OR 'A'.

Query:

```
select * from emp where ename like 'S%' or ename like 'A%';
```

✓ CORRECT

-- COMMENT: The final query you executed correctly checks both name patterns.

6. LIST ALL THE EMPLOYEES EXCEPT THOSE WHO ARE WORKING IN DEPT 10 & 20.

Query:

```
Select * from emp where deptno not in (10,20);
```

✓ CORRECT

-- COMMENT: The query correctly excludes departments 10 and 20.

7. LIST THE EMPLOYEES WHOSE NAME DOES NOT START WITH 'S'.

Query:

```
Select * from emp where ename not like 'S%';
```

✗ WRONG / ISSUE

-- COMMENT: COMMENT: The query uses curly/smart quotes ''. MySQL requires normal single quotes '. The condition itself is logically correct.

8. LIST ALL THE EMPLOYEES WHO ARE HAVING REPORTING MANAGER IN DEPTNO 10.

Query:

```
Select * from emp where mgr IS
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The query is incomplete. It ends after IS, so MySQL cannot execute it. Also, the question requires checking the department of the reporting manager, not merely whether MGR has a value.

9. LIST ALL THE EMPLOYEES WHOSE COMMISSION IS NULL AND WORKING AS CLERK.

Query:

```
Select * from emp where comm is null and job in ('CLERK');
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The query uses curly/smart quotes ' ' around CLERK. MySQL requires normal single quotes. The logical condition itself is correct.

10. LIST ALL THE EMPLOYEES WHO DON'T HAVE A REPORTING MANAGER IN DEPTNO 10 OR 30.

Query:

```
Select * from emp where deptno not in (10,30);
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The query only checks department number. It does not check whether the employee has a reporting manager, so it does not fully satisfy the question.

11. LIST ALL THE SALESMAN IN DEPT 30 WITH SAL MORE THAN 2450.

Query:

```
Select * from emp where job in ('SALESMAN') AND deptno in(30) AND sal>2450;
```

✓ CORRECT

-- **COMMENT:** The query correctly applies all three conditions. The empty result is expected because no SALESMAN in department 30 has salary greater than 2450.

12. LIST ALL THE ANALYST IN DEPT NUMBER 20 AND HAVING SALARY GREATER THAN 2500.

Query:

```
Select * from emp where job in ('ANALYST') and deptno in(20) and sal>2500;
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The query uses curly/smart quotes around ANALYST. MySQL requires normal single quotes. Your later executed version using 'ANALYST' is correct.

13. LIST ALL THE EMPLOYEES WHOSE NAME STARTS WITH 'M' OR 'J'.

Query:

```
Select * from emp where ename like 'M%' OR ename like 'J%';
```

✓ CORRECT

-- **COMMENT:** The query correctly checks both name patterns.

14. LIST ALL THE EMPLOYEES WITH ANNUAL SALARY EXCEPT THOSE WHO ARE WORKING IN DEPT 30.

Query:

```
Select *,sal*12 as annual_salary from emp where deptno not in (30);
```

✓ CORRECT

-- **COMMENT:** The query excludes department 30 and calculates annual salary using sal * 12.

15. LIST THE EMPLOYEES WHOSE NAME DOES NOT END WITH 'ES' OR 'R'.

Query:

```
Select * from emp where ename not like '%ES' AND ename not like '%R';
```

✓ CORRECT

-- **COMMENT:** The AND condition correctly requires both conditions to be true. Your earlier OR version was logically incorrect.

16. LIST ALL THE EMPLOYEES WHO ARE HAVING REPORTING MANAGERS IN DEPT NO 10 ALONG WITH 10% HIKE IN SALARY.

Query:

```
select *,sal*1.10 as hike from emp where job in ('MANAGER') AND deptno IN(10);
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The query selects employees whose own JOB is MANAGER and whose own DEPTNO is 10. The question asks about employees whose reporting manager belongs to department 10. It therefore checks the wrong employee attributes.

17. DISPLAY ALL THE EMPLOYEE WHO ARE 'SALESMAN'S' HAVING 'E' AS THE LAST BUT ONE CHARACTER IN ENAME BUT SALARY HAVING EXACTLY 4 CHARACTER.

Query:

```
Select * from emp where job in ('SALESMAN') AND ename like '%E' AND sal like '____';
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: There are multiple issues: (1) curly/smart quotes are used; (2) ename like '%E' checks E as the LAST character, not the last-but-one character; (3) SAL is DECIMAL, so LIKE '____' does not represent exactly four salary digits as intended.

18. DISPLAY ALL THE EMPLOYEE WHO ARE JOINED AFTER YEAR 81.

Query:

```
Select * from emp where hiredate>'1981-12-31';
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The written query uses curly/smart quotes around the date. MySQL requires normal single quotes. Your executed version with '1981-12-31' is correct.

19. DISPLAY ALL THE EMPLOYEE WHO ARE JOINED IN FEB.

Query:

```
Select * from emp where hiredate like '%-02-%';
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The written query uses curly/smart quotes. MySQL requires normal single quotes. Your executed version with '%-02-%' is correct.

20. LIST THE EMPLOYEES WHO ARE NOT WORKING AS MANAGERS AND CLERKS IN DEPT 10 AND 20 WITH A SALARY IN RANGE OF 1000 TO 3000.

Query:

```
Select * from emp where job not in('MANAGER','CLERK') and deptno in(10,20) and sal between 1000 AND 3000;
```

X WRONG / ISSUE

-- **COMMENT:** COMMENT: The written query uses curly/smart quotes around MANAGER and CLERK. MySQL requires normal single quotes. Your executed version using 'MANAGER','CLERK' is correct.